$$\bar{x} = \frac{\sum_{i=1}^{m} x_i}{m} = \frac{666}{9} = 74$$

$$\bar{y} = \frac{\sum_{i=1}^{n} y_i}{n} = \frac{693}{9} = 77$$

$$s_X^2 = \frac{1}{9-1} \times 628 = \frac{1}{8} \times 628 = 78.5$$

$$s_Y^2 = \frac{1}{9-1} \times 1024 = \frac{1}{8} \times 1024 = 128$$

$$F_0 = \frac{78.5}{128} = 0.613$$

Step 5 : Level of significance

$\alpha = 5\%$

Step 6 : Critical value

$$f_{(9\text{-}1,9\text{-}1),0.95} = \frac{1}{f_{(8,8),0.05}} = \frac{1}{3.44} = 0.291$$

Step 7 : Decision

Since $F_0 = 0.613 > f_{(8,8),0.95} = 0.291$, the null hypothesis is not rejected and we conclude that in school B there seems to be more variance present than in school A.

3.3 ANALYSIS OF VARIANCE (ANOVA)

In chapter 2, testing equality means of two normal populations based on independent small samples was discussed. When the number of populations is more than 2, those methods cannot be applied.

ANOVA is used when we want to test the equality of means of more than two populations. For example, through ANOVA, one may compare the average yield of several varieties of a crop or average mileages of different brands of cars.

ANOVA cannot be used in all situations and for all types of variables. It is based on certain assumptions, and they are listed below:

1. The observations follow normal distribution.
2. The samples are independent.
3. The population variances are equal and unknown.

According to R.A. Fisher ANOVA is the "Separation of variance, ascribable to one group of causes from the variance ascribable to other groups".

The data may be classified with respect to different levels of a single factor/or different levels of two factors.

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